

L2: Coordinate Transformations applied to Geodetic Sensors

Combination of measurements of stations into single coordinate system

- Local transformation between sensors (registration)

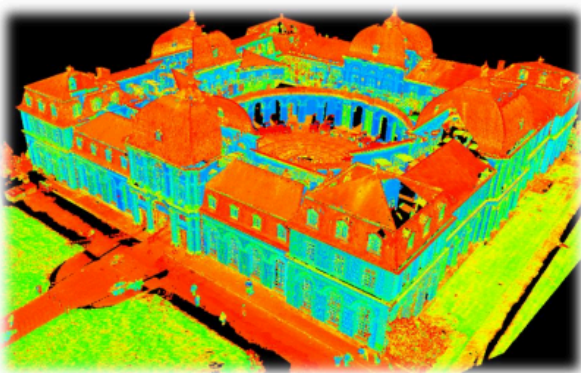
Terrestrial laser scanner (TLS)

- Need for Coordinate Transformation

Step 1: TLS Registration

- Merging to one single point cloud $\Leftrightarrow$ Coordinate transformations
- **Registration** to local coordinate system x, y, z from different local CS
 - arbitrary local CS
 - Mathematical approximation

Point cloud



3D model



Step 2: Positioning on Earth

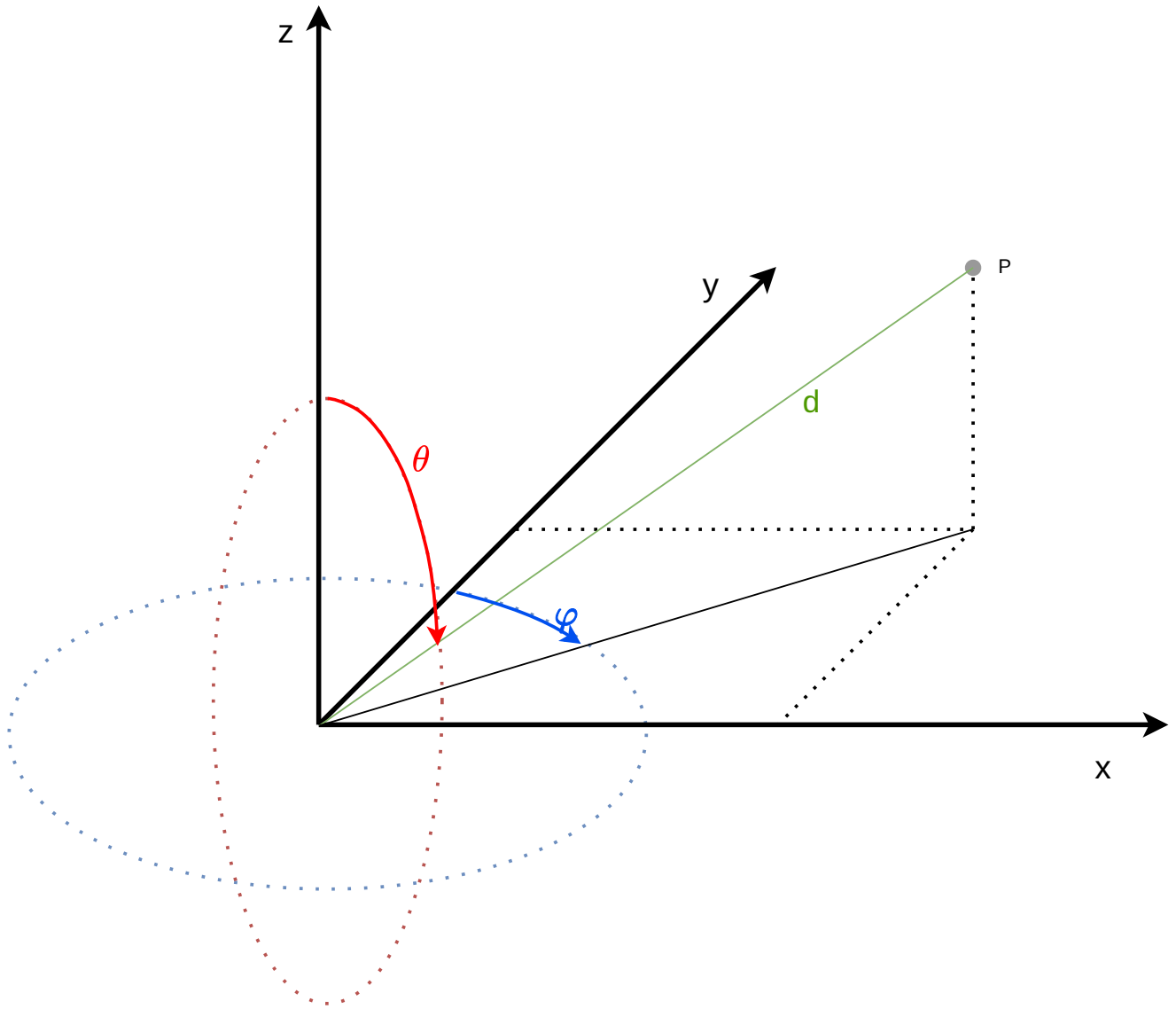
- Transformation to global, earth centered CS
- E.g. ITRF or WGS84
- **Geo-referencing**

Step 3: Projections to Given Official Coordinates

- Official Coordinates split in position (2D) and height (1D)
- Position is (UTM-)projection due to curved earth
- **Projections**

Sensor CS Realisations

- GNSS Receivers
 - Observables: x, y, z
 - CS: Geocentric, ITRF2014/WGS84
- Camera (Photogrammetry, Structure-from-Motion)
 - Observables: x, y
 - CS: local, image coordinate system
 - Coordinate transformation for 3D reconstruction
 - (Bundle adjustment)
- Total stations
 - Observables: θ, φ, d
 - CS: local, intersection of vertical, horizontal and collimation axis
 - $$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d \cdot \sin \theta \cdot \sin \varphi \\ d \cdot \sin \theta \cdot \cos \varphi \\ d \cdot \cos \theta \end{bmatrix}$$
- 3D Terrestrial Laser Scanner
 - Observables: $\theta, \varphi, d \rightarrow x, y, z$
 - CS: local, intersection of vertical, horizontal and collimation axis



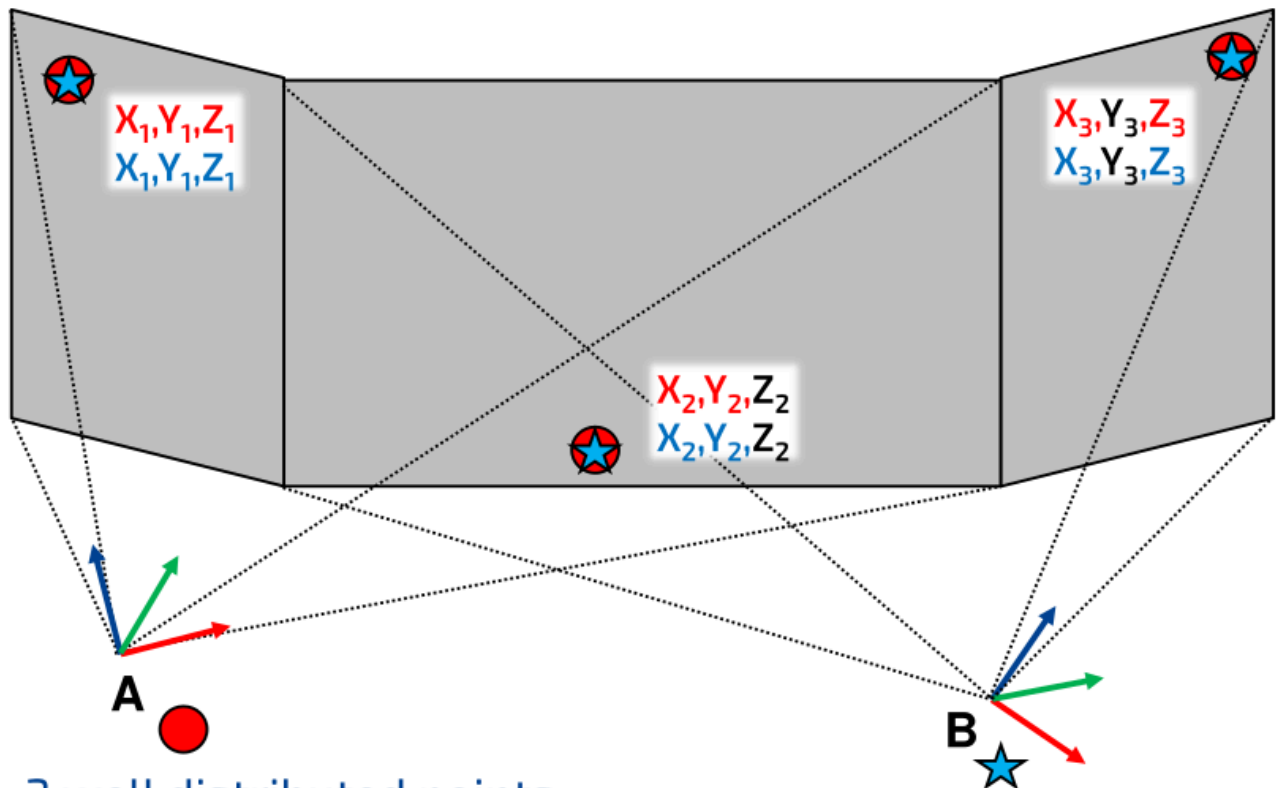
Registration

Similarity Transform in 3D:

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \mu \mathbf{R}_3(\varepsilon_z) \mathbf{R}_2(\varepsilon_y) \mathbf{R}_1(\varepsilon_x) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

- **7 DoF/ 7 parameters**
 - Scale Change (1DoF)
 - Rotation (3 DoF)
 - Translation (3DoF)
- **Determination of these 7 parameters**
- 7 or more **observations**(coordinates) in each system
 - 7 = analytical solution
 - > 7 = adjustment

- 3 Well distributed points (otherwise ill posed problem)
- known correspondences



- 3 well distributed points
- 7 or more observations (coordinates) in each system

1. Artificial targets (e.g. prism, target, sphere)

- aimed (e.g. total station) or automatically detected (TLS)

2. Well- defined keypoints (e.g. corners)

- aimed (e.g. total station) or automatically detected (TLS)

3. Point Clouds

- Most often solved by ICP

4. Geometrical features

- Varying in all 3 dimensions
- modified transformation equation

5. Alternative:

- Directly observing transformation parameters with additional hardware/sensors
 - Most often used for partial solution (some parameters) - reducing DoF
 - **Reducing number of parameters** increases efficiency and accuracy of result

Reducing DoF

- Time-of-Flight (ToF) for Scale μ

- EDM (laser): Distance given in meter -> Scale fixed
- Propagation delay of current atmosphere needs to be known
- Inclinometer for Rotations $\varepsilon_x, \varepsilon_y$
 - If reference system levelled, $\varepsilon_x, \varepsilon_y$ equal inclinations
 - Inclinometer measures: deviations from horizontal plane
 - 3D Sensors often contain compensator
 - Compensator = Inclinometer + unit correcting θ, φ
 - $\varepsilon_x = \varepsilon_y = 0$ if using only sensors with compensator

From 7 to 6 or 4 DoF

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda \mathbf{R}_3(\varepsilon_z) \mathbf{R}_2(\varepsilon_y) \mathbf{R}_1(\varepsilon_x) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

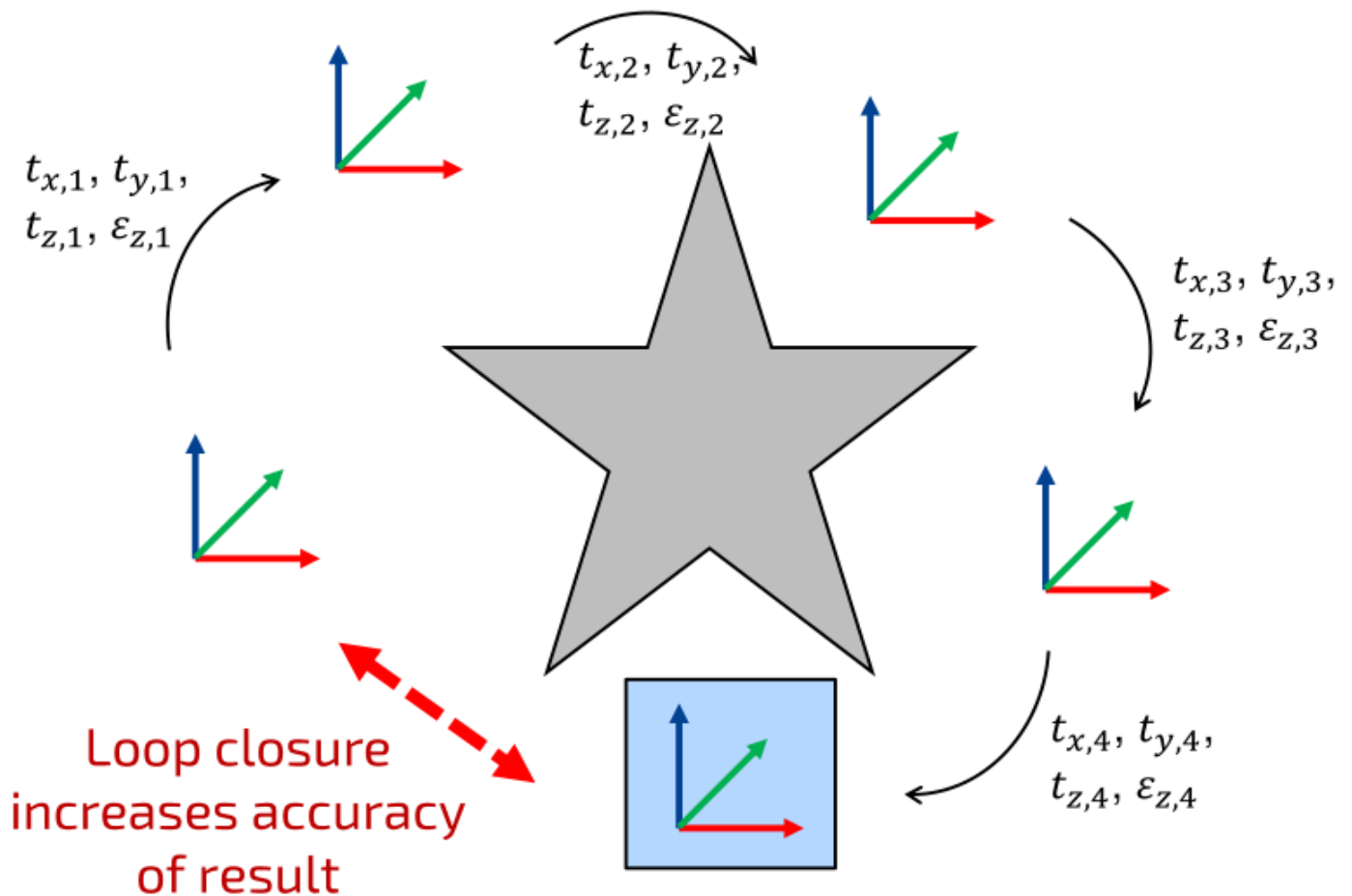
**Most
times**

$$\lambda = 1; \varepsilon_y = 0; \varepsilon_x = 0$$

**Often at static
data acquisition**

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \mathbf{R}_3(\varepsilon_z) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Composition of registrations



Summary

- Using identical artificial points (targets)
- Using measured objects (keypoints, point cloud, geometric features)
- Using additional hardware
- Combination

Geo-Referencing

- Different realisations of global coordinate systems
- Center: Center of certain ellipsoid
- X-axis: Greenwich
- Z-axis: Completion of RHS CS

-> Similarity transformation without Scale -> 6 parameters

Determination of parameters

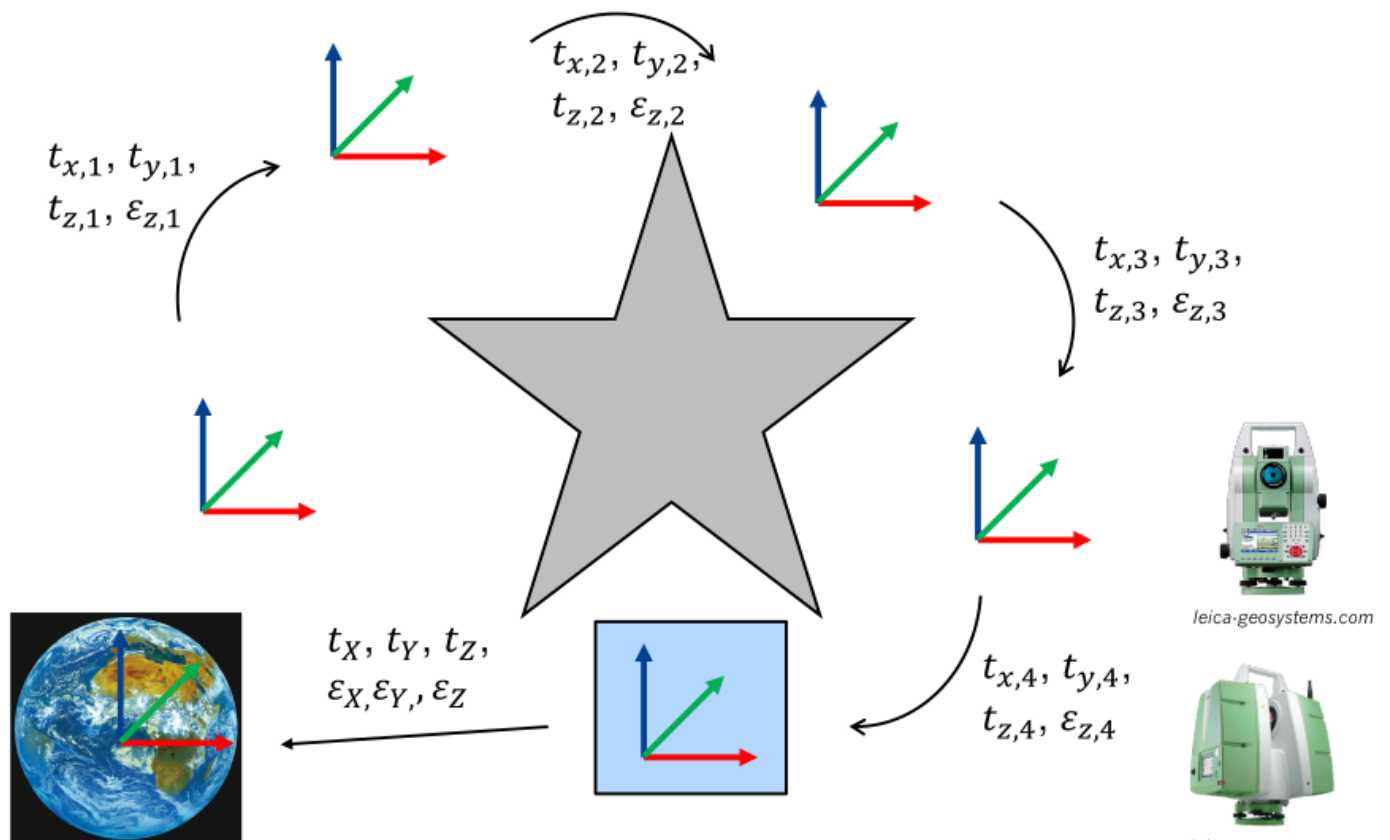
- Task for positioning on earth: Transformation to global, earth centered CS
- Determination of parameters:

1. Using identical, artificial points, **that are already given in global CS**

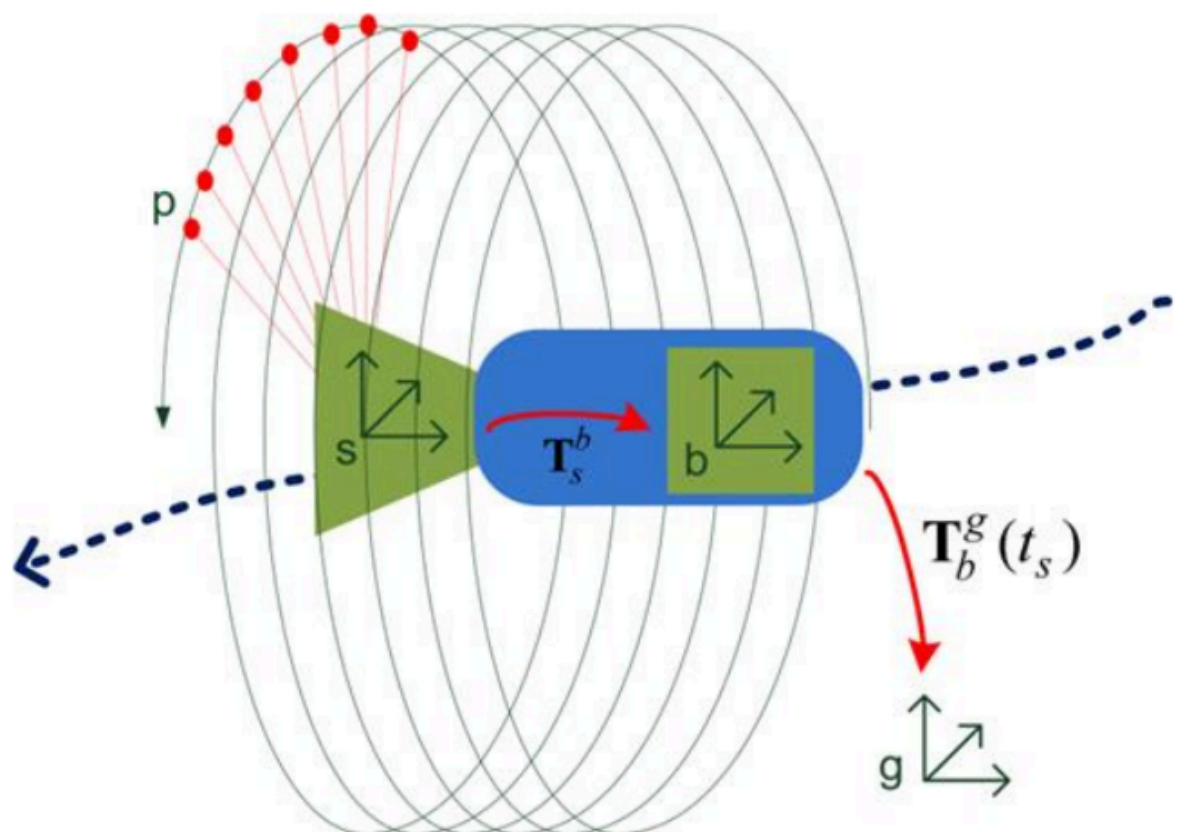
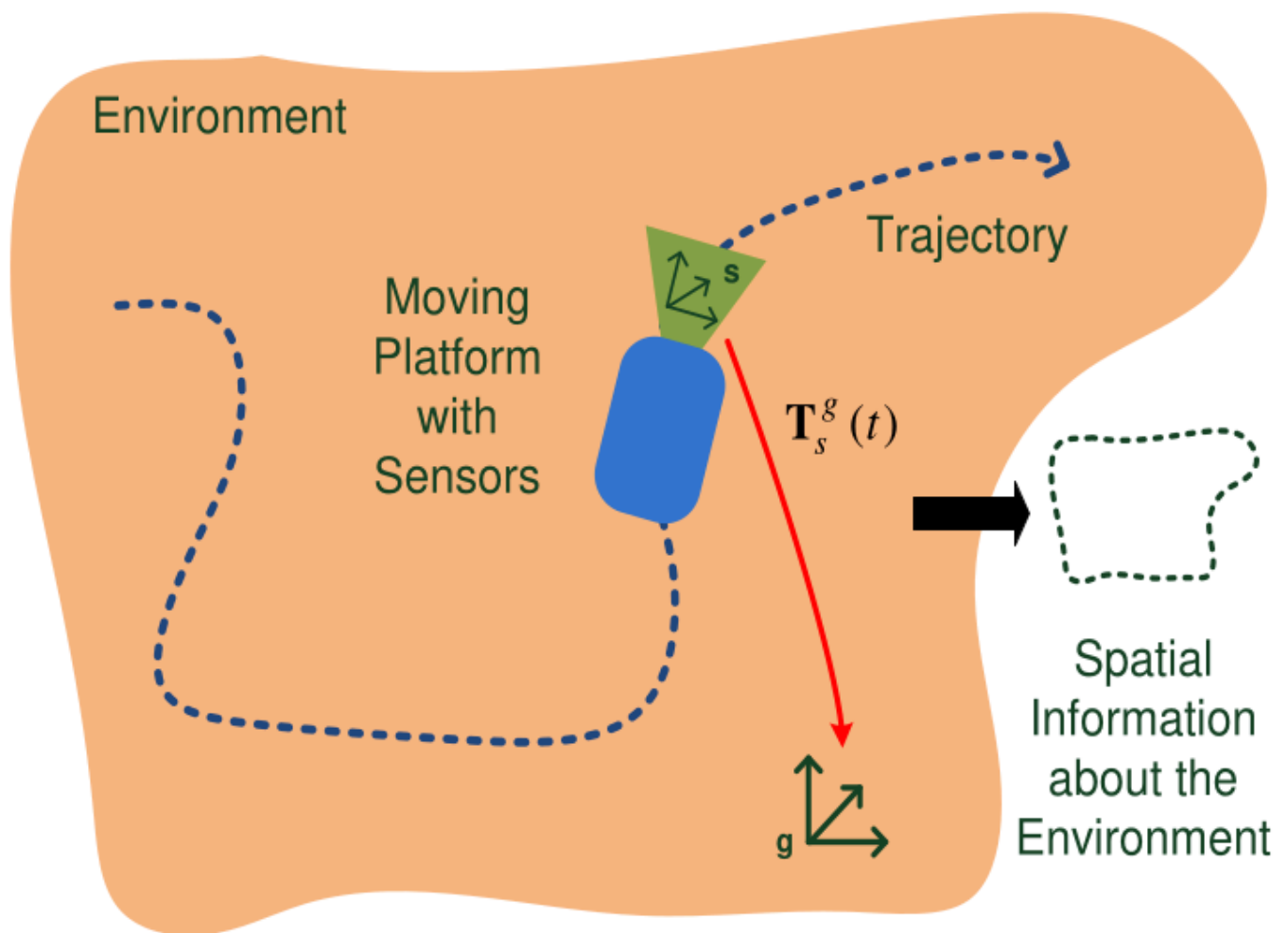
2. Using additional hardware **that acquires data in global CS**

- GNSS for t_x, t_y, t_z (most times)
- Two GNSS for ε_z
- Inclinometers for $\varepsilon_x, \varepsilon_y$
- Compass for ε_z
- Static or kinematic

Static Geo-Referencing



Mobile Multi-Sensor Systems



- Geo-referencing by: $\mathbf{p}^g = \mathbf{T}_b^g(t_s)\mathbf{T}_s^b\mathbf{p}^s$